INCH-POUND

MIL-W-0052960B(ME)
24 June 1994
USED-IN-LIEU-OF
MIL-W-52960A
12 December 1984

MILITARY SPECIFICATION

WATER PURIFICATION UNIT, REVERSE OSMOSIS,

600 GPH

This specification is approved for use by the US Army Belvoir Research, Development, and Engineering Center based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-W-52960A. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This specification covers a military design reverse osmosis water purification unit (ROWPU) having a nominal capacity of 600 gallons per hour (gph) on fresh water.

1.2 Classification. The water purification units shall be of the following types, as specified (see 6.2.1):

Type I - Trailer mounted.
Type II - Skid mounted: two fork pocket.
Type III - Skid mounted: four fork pocket.

Beneficial comments (recommendations, additions, deletion) and any pertinent data which may be of use in improving this document should be addressed to: us ARMY BELVOIR RDE CTR, ATTN SATBE TSE, 10101 GRIDLEY RD, STE 104, FT BELVOIR VA 22060-5818 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 4610

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards ((DoDISS) and supplement thereto cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated Plywood.
PPP-B-636	Boxes, Shipping, Fiberboard.
PPP-B-1055	Barrier Material, Waterproofed, Flexible.
PPP-P-40	Packaging and Packing of Hand Tools.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing).

MILITARY

MIL-C-104	Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	Preservation-Packaging, Methods of.
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173	Varnish, Moisture- and Fungus-Resistant (for Treatment of Communication, Electronic, and Associated Equipment) .
MIL-P-514	Plates, Identification, Instruction, and Marking, Blank.
MIL-T-704	Treatment and Painting of Materiel.
MIL-F-3541	Fittings, Lubrication.
MIL-G-3859	Grease Guns, Hand-Operated, Lever, Push, and Screw Type.
MIL-W-6858	Welding, Resistance: Aluminum, Magnesium, Nonhardening Steels or Alloys, Nickel Alloys, Heat-Resisting Alloys, and Titanium Alloy; Spot and Seam.
MIL-B-7883	Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys .
MIL-P-10603	Pumps and Pumping Units, Centrifugal, Power-Driven, for Water, Packaging of.
MIL-G-10924	Grease, Automotive and Artillery.
MIL-P-19834	Plates, Identification or Instruction, Metal Foil, Adhesive Backed, General Specification for.

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MIL-G-23827 - Grease, Aircraft and Instrument, Gear Actuator Screw.
MIL-V-62038 - Vehicle, Wheeled, Preparation for Shipment and Storage of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1366 - Materiel Transportation System Dimensional and Weight Constraints, Definition of.
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipments, and Facilities.
MIL-STD-1474 - Noise Limits for Army Materiel.
MIL-STD-1791 - Designing for Internal Aerial Delivery in Fixed Wing Aircraft.
MS35844 - Plug, Machine Thread, Magnetic: Drain.
MS49006 - Plugs, Pipe, Magnetic-Headless: Aluminum, Iron or Steel.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: STDZN DCMNT ORDER DESK, BLDG 4D, 700 ROBBINS AVE, PHILADELPHIA PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in effect on the date of the solicitation.

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DRAWINGS

ME

TA 13222E5235	Water Purification Unit, Reverse Osmosis, Trailer Mounted, 600 GPH.
TA 13225E8940	Water Purification Unit, Reverse Osmosis, Skid Mounted, Four Fork Pocket, 600 GPH.
TA 13226E0121	Water Purification Unit, Reverse Osmosis, Skid Mounted, Two Fork Pocket, 600 GPH.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from: USA BELVOIR RDE CTR, ATTN SATBE FSQ, 10101 GRIDLEY RD, STE 104, FT BELVOIR VA 22060-5818.)

PUBLICATIONS

TM5-4610-215-10 - Water Purification Unit, Reverse Osmosis, 600 GPH, Trailer Mounted, Flat Bed Cargo, 5 Ton, 4 Wheel Tandem (Model ROWPU 600-1).

CODE OF FEDERAL REGULATIONS (CFR)

TITLE 21- Federal Food, Drug and Cosmetic, Food Additives.

(Application for copies of the publications should be addressed to: SUPT OF DCMNTS, GVT PRINTG OFC, WASHINGTON DC 20402.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1141	- Substitute Ocean Water.
D 3951	- Standard Practices for Commercial Packaging.
D 3953	- Standard Specification for Strapping, Flat Steel and Seals.

(Application for copies should be addressed to: AMERCN SCTY FOR TEST & MTRLS, 1916 RACE STRET, PHILADELPHIA PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

A5.1 -	Mild Steel Covered Arc-Welding Electrodes.
A5.6 -	Covered Copper and Copper Alloy Arc Welding Electrodes, Specification for.

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A5.7 - Copper and Copper Alloy Bare Welding Rods and Electrodes, Specification for.
A5.9 - Corrosion-Resisting Chromium and Chromium-Nickel Steel Welding Rods and Bare Electrodes.
A5.10 - Aluminum and Aluminum-Alloy Welding Rods and Bare Electrodes.

(Application for copies should be addressed to: AMERCN WELD SCTY, 550 NW LEJEUNE RD, PO BOX 351040, MIAMI FL 33135.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J429 - Mechanical and Material Requirements for Externally, Threaded Fasteners.
J492 - Rivets and Riveting.

(Application for copies should be addressed to: SCTY OF AUTOMTV ENGRS, INC, 400 CMNWLTH DR, WARRENDALE PA 15096.)

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to: S MGMT CNTRL DIST, 9150 E FLAIR DR, EL MONTE CA 91731.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS drawings), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The water purification units shall be in accordance with top assembly TA 13222E5235 for type I, top assembly TA 13226E0121 for type II, top assembly TA 13225E8940 for type III, and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval from the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to

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ensure correct fit, assembly, and operation of the water purification unit. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.4).

3.3 Materials.

3.3.1 Material deterioration prevention and control. The water purification unit shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the water purification unit may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Accectable materials.

3.3.3.1 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waster and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces, and parts incorporated in the water purification unit may be newly fabricated from recovered materials to the maximum extent practicable, provided the water purification unit produced meets all other requirements of this specification. Used, rebuilt, or remanufactured components, pieces, and parts shall not be incorporated in the water purification unit.

3.3.3.2 Safe materials. Material shall be as specified herein and on the applicable drawings (see 6.13) . Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. The material shall have no adverse effect on the health of personnel when used for its intended purpose. The lines and all surfaces that contact potable water shall conform to the Federal Food, Drug and Cosmetic, Food Additives, Title 21, Code of Federal Regulations for contact with food. The compounds listed in sections 177.1020, 177.1030, 177.1040, 177.1050, and 177.1480 of the Federal Food, Drug and Cosmetic, Indirect Food Additives (Chapter

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1, Title 21, Code of Federal Regulations) , shall not be acceptable materials. Questions pertaining to this effort shall be referred by the contracting officer to the appropriate departmental medical service, which will act as an advisor to the contracting agency (see 6.5) .

3.4 Operation. The water purification unit shall operate at design capacity without leaks, malfunctions, or permanent deformation. The water purification unit, when operated in accordance with the applicable technical manual, shall have a water production capacity of at least 600 gph on fresh water containing less than 500 ppm of total dissolved solids at a water temperature of 77 “F. When tested as specified in 4.6.2.9, the production rate for sea water shall be at least 400 gph.

3.4.1 Total dissolved salts rejection. When tested as specified in 4.6.2.3 and 4.6.2.9.2, the water purification unit shall produce product water containing a maximum of 1,500 ppm dissolved solids.

3.5 Environmental.

3.5.1 Operating temperature. When tested as specified in 4.6.2.5, the water purification unit shall perform as specified in any ambient temperature from -110 to -25 °F. Shelter is to be used when the temperature is below 32 ‘F to provide the required protection for operation down to -25 “F.

3.5.2 Storage temperature. When tested as specified in 4.6.2.5, the water purification unit shall not be damaged by storage at ambient temperatures from +160 to -30 “F. The reverse osmosis elements and calcium hypochlorite shall be removed from the unit during storage.

3.6 Safety. The water purification unit shall not present any safety hazard to operating and maintenance personnel and shall include, but not be limited to, the following general safeguarding provisions:

a. All reciprocation, rotating, or moving parts of the equipment shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard.
b. All points requiring lubrication during operation shall have fittings located or guarded so as to be accessible without hazardous exposure to personnel performing those operations.
c. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard.
d. Nonfunctional sharp edges, projecting points, and excessive length of fastening devices shall be avoided.

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e. Access to the chemical feeder shall not cause splashing of chemicals.
f. Each container of chemicals (citric acid, polyelectrolyte, surfactant, or sodium hexametaphosphate) shall be labelled with the generic name and the following note:

"CAUTION: Protect eyes and skin from direct contact with chemicals. Be especially careful when opening containers to avoid inhaling chemical powder. Wash hands thoroughly in soap and water after handling chemicals.?'

Labels shall be on identification plates conforming to MIL-P-19834, type 1, style 2.

3.7 Human factors. The water purification unit shall conform to the applicable human factors engineering criteria as specified in MIL-STD-1472 where practicable. The water purification unit shall be operable and maintainable by the 5th to 95th percentile.

3.7.1 Noise level. When tested in accordance with 4.6.2.8, the noise level of the water purification unit (excluding horns and warning slams) shall be in accordance with MIL-STD-1474, with the exception of 5.2, 5.3, and 5.4. The noise level at the operator's station and occasionally occupied positions shall not exceed Category D of MIL-STD-1474. The operator's position is defined as 18 inches from the control panel. Occasionally occupied positions are defined as anywhere within 39 inches of the water purification unit except for the operator's position. Where the limit of Category D can be documented as being clearly beyond the state-of-the-art, as described in 5.1.1.2 of MIL-STD-1474 selection of another noise limit shall be made by the procuring activity and the requirements of 5.1.1.3 shall apply.

3.8 Reliability. When tested as specified in 4.6.2.9, the specified mean-time-between-failure (MTBF) of the water purification unit and ancillary equipment (not including the generator) shall be 500 hours. During reliability testing, the trailer shall exhibit no failures.

3.9 Maintainability. The water purification unit shall have a maintenance ratio of not more than 0.03. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hours for repair of replaced components and scheduled before- and after-operation checks are excluded. A maintenance schedule shall be furnished prior to the start of any testing. The generator shall not be included in the maintenance ratio.

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3.10 Electromagnetic interference. The electromagnetic interference emission characteristics of the water purification unit shall conform to MIL-STD-461, Class Cl, Tactical and Special Purpose Vehicles and Engine-Driven Equipment.

3.11 Trailer electrical system. All lights and electrical circuits shall be capable of operation throughout all vehicle operating conditions.

3.12 Trailer brake system. The trailer shall be equipped with an automatically engaging service brake which allows the brakes to engage when the hoses between the ROWPU and its prime mover are disconnected. The service brake shall maintain application of brakes for not less than 15 minutes when hoses are disconnected.

3.13 Levelling jacks. The jacks shall have the capacity to raise and support the ROWPU.

3.14 Lubrication.

3.14.1 Lubricants. The water purification unit shall operate as specified herein when serviced with military lubricants (see 6.7).

3.14.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, type I, II, or III. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun conforming to MIL-G-3859 with a 10-inch flexible extension. Accessibility of fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

3.14.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment could damage grease seals or other parts.

3.14.4 Filling, draining, and checking Provisions. Enclosures such as gear cases and transmission housings that contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, check plugs not less than l/2-inch pipe size, or sight gauges to determine the level of the lubricant. Each enclosure shall be equipped for filling with a lubricant and for draining. The drain outlet shall be fitted with a magnetic drain plug, size conforming to MS35844 or MS49006. The drain outlet shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be accomplished when the water purification unit is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain outlet. ”

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3.14.5 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be with greases as follows:

a. MIL-G-23827 - For ball, roller, and needle bearings, gears, and on sliding and rolling surfaces of such equipment as instruments, cameras, electronic gear and mechanical control systems; for use on gears, actuator screws, and other equipment requiring a lubricant with high load-carrying capacity over a temperature of -100 to +250 “F.

b. MIL-G-10924 - For lubrication and surface corrosion protection of automotive and artillery equipment operated over the temperature range of -65 to +225 ‘F. Also used in other applications within this temperature range where grease with oxidation-resistant and corrosion-preventive properties is desirable. Not for use on machinery which comes in contact with foods or potable water.

The water purification unit shall be assembled, broken-in, tested, and delivered with these greases. The bearings shall be cleaned before lubrication because military greases are not always compatible with other greases. A tag shall be attached in a conspicuous place to indicate which military grease has been used.

3.14.6 Oil lubrication. Unless otherwise preserved for packaging (see 5.1), and upon completion of all testing, all components shall be filled to the operating level with winter grades of military oils designed for use in the temperature range of 0 to +32 ‘F, A tag shall be attached in a conspicuous place to indicate the temperature range and the lubricant grade used.

3.15 Fungus and moisture resistance. The electrical circuitry, including all components and connection, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition as specified in 3.15.1, with 1 percent by weight copper 8-quinolinolate based on the nonvolatile component of the varnish. Components that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated. When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.15.1 Composition. Composition I shall be used unless composition II is required in order to comply with local air

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pollution regulations concerning the application of the varnish. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required and shall certify that the composition II material complies with Rule 442, South Coast Air Quality Management District.

3.16 Identification marking. The water purification unit shall be identified in accordance with MIL-STD-130. The marking shall be applied to the water purification unit on plates conforming to MIL-P-514, type III, composition C, type I or II, class 1 material. Information included shall be as specified on the appropriate drawing. The plates shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

3.17 Shipping data plate. A shipping data plate conforming to MIL-P-514, type III, composition C, type I, grade A, class 1 material shall be provided. In addition, the plate shall show the silhouette of the water purification unit in transport position indicating the center of gravity and the location and capacity of the slinging and tie-down provisions. The plates shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

3.18 Instruction Plates. Each water purification unit shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the water purification unit. Instruction plates shall conform to MIL-P-514, type III, composition C, type I, grade A, class 1 material. The plates shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

3.19 Treatment and painting. Treatment, painting, and color shall be in accordance with MIL-T-704 or the applicable drawing. Color shall be as specified by the contracting officer (see 6.2).

3.20 Transportability. The water purification unit shall be capable of being transported by military or commercial trailers, trains, marine vessels, and aircraft, and must be capable Of withstanding the impact forces encountered in shipment without damage or permanent deformation. The water purification unit shall be equipped with tiedown or slinging provisions.

3.20.1 Tiedown Provisions. The tiedown provisions shall conform to MIL-STD-209, class 2 or 3, type IV equipment, and to MIL-STD-1791 for equipment restraint criteria. The tiedown provisions shall satisfactorily complete the pull testing specified in 4.6.2.12.1 without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or

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its connecting structural components. Tiedown points shall be labelled “TIE DOWN HERE". Combination lift/tiedown points shall be labelled "LIFT/TIE DOWN HERE”. The labels shall conform to MIL-STD-1472.

3.20.2 Slinging Provisions. The slinging provisions shall conform to MIL-STD-209, class 1 or 3, type IV equipment. The provisions shall enable the complete water purification unit to be lifted in the normal operating position. The provisions shall be located so that not less than one inch clearance is maintained between slings and all exterior parts and shall be fastened to members which will withstand stresses in the amount and direction of pull specified for the provisions without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural components when tested as specified in 4.6.12.2. Slinging provisions may also be used a tiedown provisions when such provisions meet the requirements of 3.20.1. Lift points shall be labelled “LIFT HERE”.

3.20.3 Air transportability and airdrop. When tested in accordance with 4.6.2.12.3, the water purification unit shall meet the requirements of MIL-STD-1791 for air transport and air drop. The water purification unit shall be transportable by C-130 or C-141 aircraft and lifted by medium-lift helicopter.

3.20.4 Rail transportability. The water purification unit shall be rail transportable in CONUS and NATO countries without restrictions. When loaded on a 50 inch (127 centimeter) high rail car, the water purification unit shall have a dimensional profile within the AAR outline diagram and the Gabarit International de Chargement or PPI, as specified in MIL-STD-1366. When tested as specified in 4.6.2.12.4, the trailer mounted water purification unit shall in addition, withstand rail impact without damage or deformation.

3.20.5 Highway transportability. When tested as specified in 4.6.2.12, the water purification unit shall withstand shock and vibration without damage or permanent deformation to any part or component. In addition, when loaded on a semitrailer/tractor it shall be within the highway permit limits of all states.

3.21 Government-furnished property. One operator’s manual, TM5-4610-215-10, for each water purification unit, will be furnished by the Government (see 6.14).

3.22 Vehicle weight classification. The contractor shall complete the Government-furnished vehicle weight classification forms and apply the vehicle weight classification number to the water purification unit as requested (see 6.8) .

3.23 Workmanship. All parts, components, and assemblies of the water purification unit including castings, forgings, molded

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parts, stampings, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. Al 1 edges shall be rounded or chamfered. The unit shall be free from such defects as misaligned components, incomplete welds, rust, extraneous material, cracks, leaks, and other defects that could impair the operation and accessibility of the unit.

3.23.1 Metal fabrication. Metal used in the fabrication shall be free from kinks and sharp bends. The material shall be straightened by methods that will not injure it. Corners shall be square and true. Flame-cutting, using tips suitable to the thickness of the steel, may be used instead of shearing and sawing. All bends shall be made with controlled means to ensure uniformity of size and shape. Precautions shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.23.2 Bolted connections. Bolt holes shall be accurately formed and shall have the burrs removed. Washers or lock washers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torque and shall have full thread engagement.

3.23.3 Riveted connections. Rivets shall fill the holes completely. All rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member and shall be in accordance with SAE J492.

3.23.4 Welders, weldments, and welding.

3.23.4.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall develop a certification of welder qualification for the welder (see 6.3).

3.23.4.2 Weld filler metals. Aluminum filler metal shall be in accordance with AWS A5.10, type ER5356 or ER5556 or ER4043. Steel filler metal shall be in accordance with AWS A5.1, type E7018 or E6011 for carbon steel, type E7018 for high-strength low alloy steel, type E6018 for high-strength low alloy steel to carbon steel, and AWS A5.9, type ER308L for corrosion resisting steel. Copper-nickel alloy filler material shall be in accordance with AWS A5.6 or A5.7.

3.23.4.3 Aluminum, copper-nickel, and steel weldments. The surfaces and edges of parts to be welded, including each layer of weld materials when multiple layers are requires, shall be free

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from oxide scale, paint, lubricant, and other foreign matter. Welds shall transit stress to the heat-affected zone without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. All welded parts shall be free from cracks, porosities, and other imperfections that may reduce the effectiveness of the part. Welds involving corrosion-resistant steel shall be passivated and annealed as specified on the applicable drawings.

3.23.4.3.1 Aluminum welding. Aluminum welding, except spot and seam welding, shall be done by the metal inert gas (MIG) method or other methods that will provide equivalent mechanical properties of the filler metal to base metal combinations specified in 3.23.4.2. All welds shall be made at a uniform rate of speed. Complete and uniform penetration and fusion of the metals shall be obtained on all welds without metallic or nonmetallic inclusions. Finished aluminum welds may be ground, filed, chipped, or wire-brushed, but shall not be hammered. Preheating for welding is permissible provided the temperature does not exceed 300 “F for a total time not to exceed 30 minutes. No process requiring welding flux shall be used. Work shall be positioned and rigged for flat welding whenever practicable.

3.23.4.3.2 Steel welding. Steel welding shall be done by the metal inert gas (MIG) method in accordance with the qualification procedures prescribed in 3.23.4.1, or other methods that will provide equivalent mechanical properties of the filler metal to base metal combinations specified in 3.23.4.2. Where embrittlement of the joined metal may result the work shall be preheated or annealed or both. Work shall be positioned for flat welding whenever practicable. Complete and uniform penetration and fusion of the metals shall be obtained on all welds.

3.23.4.3.3 Copper-nickel alloy welding. Copper-nickel alloy welding shall be done by one of the following methods: shielded metal-arc welding, gas metal arc MIG welding, or gas tungsten arc welding. Flat position welding shall be used unless permission is obtained from the contracting officer for use of horizontal, vertical, or overhead welding. Square butt joints shall not be used.

3.23.4.4 Weldment specimens. The contractor shall submit specimens representing each typical weld joint utilized in fabrication of the water purification unit. These specimens shall be furnished at the same time as the first article water purification unit. Weldment specimens shall be submitted by each welder performing weldments on the water purification unit with each welding equipment used. Acceptable specimens shall represent the minimum acceptable weld quality and shall be retained as an inspection standard for evaluation of production weldments.

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3.23.4.5 Welding procedure. Prior to any welding the contractor shall prepare and submit to the contracting officer for approval a written welding procedure defining process and control. The procedure shall include, but is not limited to, the factors listed in table I. Welding procedure changes affecting any of the factors in table I shall be submitted to the contracting officer for approval prior to adoption of the change.

3.23.4.6 Rewelding defects. Portions of welded joints that have been rejected on inspection because of defects specified in 3.23.4.3 or not conforming to the weldment specimens specified in 3.23.4.4 may be repaired only by rewelding. The defective weld shall be removed by chipping or machining. With the exception of steel, flame cutting shall not be used. Before rewelding, the joint shall be inspected to ensure that all of the defective weld has been removed and that the joint is accessible so that the welding operator can obtain full penetration and fusion through the joint. Aluminum shall not be repaired by patching.

3.23.4.7 Spot and seam welding. Spot and seam welding shall be in accordance with MIL-W-6858.

TABLE I. Factors to be included in welding procedures.

Item No	Factor
1.	Welding process and techniques including shielding medium.
2.	Welding process operation.
3.	Type of current, ac or dc; polarity of dc, if used.
4.	Shielding gas composition.
5.	Joint design.
6.	Method of joint preparation.
7.	Preweld and interpass cleaning.
8.	Preheat or ambient temperature range.
9.	Composition of filler metal.
10.	Filler metal size for all passes.
11.	Number and sequence of passes.
12 l	Welding position.
13*	Repair method to be used in removing defects.
14.	Method of inspection used to ensure proper repair.
15*	Method of qualification of welders.
16.	Method of determining capability of automatic equipment and equipment operators.
17 l	Workmanship specimens.
18.	Method of inspection of finished welds.

3.23.4.8 Brazing. Brazing of steels, copper alloys, and nickel alloys shall be in conform to MIL-B-7883.

3.23.5 Diagrams, charts, and plates. Diagrams, charts, and plates attached to components with adhesive shall have surfaces

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free of wrinkles, bubbles, or other defects that may cause them to become loose or damaged.

3.23.6 Cleaning. The water purification unit shall be thoroughly cleaned of all cutting oil, test residue, grease, dirt, metal scrap, and other contaminants. Cleaning shall be done in a way that will not leave a residue or otherwise make the water purification unit unsuitable for its intended use. Except where moisture is required for presentation during shipment, the unit shall be completely drained before preparation for packaging.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order and unless disapproved by the Government, the contractor's own facilities or any other, suitable for the performance of the inspection requirements specified herein, may be used. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and service conform to prescribed requirements.

4.1.1 Acceptability criteria. Water purification units which conform to all requirements in sections 3 and 5 of this specification and pass all applicable inspections and tests in section 4 of this specification will be considered acceptable by the Government. The Government is the sole judge of compliance with section 3, 4, and 5.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with the applicable specifications, standards, and drawings.

4.1.3 Disassembly inspection. Failure of any examination or test by the first article water purification unit may be cause for disassembly in the presence of a Government representative of the first article to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and the applicable drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first article. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.1.4 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the quality

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assurance provisions (QAPs) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the bases for acceptance and rejection (see 6.9 and 6.10.1).

4.1.5 End item final inspection requirement (EIFIR). Each water purification unit shall be inspected in accordance with the EIFIR corresponding to its top assembly drawing (e.g., type I ROWPUs shall be inspected to EF 13222E5235). One copy of the completed EIFIR shall be shipped with each inspected water purification unit (see 6.10.2).

4.1.6 Responsibility for compliance. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this specification shall not relieve the-contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection as part of manufacturing operations is an acceptable practice to ascertain conformance to requirements. This does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.5).
c. Inspection of packaging (see 4.7).

4.3 First article inspection.

4.3.1 First article examination. The first article shall be examined as specified in 4.5.1 in the order shown. Presence of one or more defects shall be cause for rejection and, at the contracting officer's discretion, for performing the disassembly inspection specified in 4.1.3.

4.3.2 First article tests. Upon successful completion of the examination in 4.3.1, the first article shall be subjected to the tests marked "X" in column 1 of table III. Failure of any test shall be cause for rejection or, at the contracting officer's discretion, for performing the disassembly inspection specified in 4.1.3.

4.4 Inspection failure. Failure of a first article water purification unit to meet any requirements specified herein during and as a result of the examination and tests specified in 4.3 shall be cause for rejection of the first article water purification unit and shall be cause for refusal by the

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Government to continue acceptance of production water purification units until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on water purification units previously produced and accepted under the contract. Any deficiencies found as a result of the first article inspection shall be considered prima facie evidence that all water purification units accepted prior to the completion of first article inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each water purification unit shall be examined for the characteristics shown in table II. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Upon successful completion of the examination in 4.5.1, each water purification unit shall be subjected to the tests marked “X” in column 2 of table III. Failure of any test shall be cause for rejection.

4.6 Inspection schedule.

4.6.1 Examination. Examination shall be in accordance with table II.

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TABLE II. Examination schedule.

No.	Defect
101.	Any part or component not in accordance with the QAP requirements as shown on the drawings (see applicable drawings) .
102.	Any dimension, other than those identified by the QAP requirements shown on the drawings, not as specified [see applicable drawings) .
103.	Assembly not complete or not correct (see applicable drawings) .
104.	Parts or components missing (see applicable drawings) .
105.	Parts or components not as specified (see applicable drawings) .
106.	Reflectors on trailer not as specified (see applicable drawings) .
107.	Air brakes not as specified (see applicable drawings) .
108.	Frame not as specified (see applicable drawings) .
109.	Electrical installation not as specified (see applicable drawings) .
110.	Lunette not as specified (see applicable drawings) .
111.	Chain assembly not as specified (see applicable drawings) .
112.	Bumper stops not as specified (see applicable drawings).
113.	Spare tire and wheel assembly not as specified (see applicable drawings) .
114.	Suspension assembly not as specified (see applicable drawings) .
115.	Swivel assembly not as specified (see applicable drawings) .
116.	Jack assembly not as specified (see applicable drawings) .
117.	Generator mounting holes not as specified (see applicable drawings) .
118.	Material not as specified (see 3.3 and applicable drawings).
119.	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.2) .
120.	Used, rebuilt or remanufactured components, pieces, or parts incorporated in the ROWPU (see 3.3.3.1).
121*	Lubrication not as specified (see 3.4).
122.	Lubricants not as specified (see 3.4.).
123.	Lubrication fittings not as specified (see 3.4.2).
124.	Pressure relief device not as specified (see 3.4.3).

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TABLE II. Examination schedule. (Continued)

No.	Defect
125.	Filling, draining, and checking provisions not as specified (see 3.4.4).
126.	Grease lubrication not as specified (see 3.4.5).
126.	Oil lubrication not as specified (see 3.4.6).
128.	Protection against fungus and moisture resistance not as specified (see 3.5).
129.	Safety not as specified (see 3.6).
130.	Chemical containers not labelled as specified [see 3.6).
131.	Identification not as specified (see 3.6).
132.	Shipping data plates missing, not attached, or illegible (see 3.7).
133.	Instruction plates missing, not attached, or illegible (see 3.8).
134.	Treatment and painting not as specified (see 3.9 and applicable drawings) .
135.	Vehicle weight classification missing (see 3.22).
136.	Workmanship not as specified (see 3.23, 3.23.1, and 3.23.5).
137.	Water purification unit not completely drained and cleaned before preparation for packaging (see 3.23 and 3.23.6).

4.6.2 Tests.

4.6.2.1 Test schedule. Tests shall be in accordance with table III.

4.6.2.2 Test conditions. Unless otherwise specified in a test, tests shall be performed without shelter and at the environmental conditions existing at the place of test. If the ambient temperature is below 32 ‘F shelter may be provided to bring the temperature up to 32 “F. The water purification unit shall operate as specified herein without maintenance other than that prescribed by the operator’s manual. Prior to the start of the test any components of the water purification unit requiring lubrication shall be lubricated with the oils and greases specified in 3.14. Oils shall be those designated-for use in the ambient temperature at the place of the test.

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TABLE III. Test schedule.

First Article	Quality Conform	Test	Reqmnt	Test Para
1	2	3	5	6
x		operational (40 hours).		
	x	operational (6 hours) .	3.4	4.6.2.4
x		Environmental.	3.5	4.6.2.5
x		Safety.	4.6	4.6.2.6
x		Human factors.	3.7	4.6.2.7
x		Noise level.	3.7.1	4.6.2.8
x		Reliability.	3.8	4.6.2.9
x		Maintainability.	3.8	4.6.2.10
x		Electromagnetic interference.	3.10	4.6.2.11
x		Transportability.	3.20	4.6.2.12
x	x	Trailer electrical system (type I only) .	3.11	4.6.2.13
x	x	Trailer brake system (type I only).	3.12	4.6.2.14
x	x	Leveling jacks (type I only).	3.13	4.6.2.15

4.6.2.3 Operational test (40 hours). When the reliability test (see 4.6.2.9) is not being performed, the first article water purification unit shall be operated as specified in 4.6.2.2, and 4.6.2.3.1 through 4.6.2.3.3 for not less than 40 hours. At least 5 back-washing cycles shall be performed during the 40-hour period. Upon completion of the test, the unit shall be flushed as specified in 4.6.2.3.2.

4.6.2.3.1 Test equipment. Test equipment shall include, but is not limited to, the following:

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Power source -	120/208 volt, 3-phase, 4-wire, 60 Hertz.
Raw water pump -	self priming, 30 gpm at 120 foot head.
Backwash pump -	centrifugal, 120 gpm at 75 foot head.
Suction hose -	cam and groove couplings.
Calcium hypochlorite -	commercial, 65 percent available chlorine, minimum.
Polyelectrolyte, cationic -	Catfloc TL or equal.
Test feedwater -	ASTM D1141, substitute ocean water, formula A, section 6 without heavy metals, shall be used, except the salt shall be dissolved in dechlorinated tap water to provide a solution containing 19,000 ±1000 mg/1 chloride ion. Sodium thiosulfate may be used to remove chlorine. (Note: Natural sea water may be substituted for the artificial ocean water with the contracting officer's approval (see 6.2.)

4.6.2.3.2 Test Procedure. The water purification unit with ancillary equipment (see 6.13) shall be set up and maintained in accordance with the applicable technical manual. The unit shall be operated for twenty hours a day during the reliability and 40-hour operational tests. The permeate and concentrate streams may be blended for recirculation provided that the temperature of the feed does not exceed 90 "F. When the water is recirculated, dechlorinated water shall be used in the chlorine chemical feed can. The other chemicals shall be used until the total volume of water to be recirculated has been processed, at which time they shall be replaced with dechlorinated water. Only tools and parts normally provided with the water purification unit, in the quantities normally provided, may be used for maintenance of the water purification unit during the test. After the test is complete, the water purification unit shall be flushed with fresh, chlorine free water for ten minutes. During the flush, the chemical feed cans shall be filled with fresh, chlorine free water. The control knobs on the chemical feed pump shall be set fully on. After the flush is completed, open and leave open all drain and vent valves. Follow the instructions in manual TM5-4610-215-10 for preparation for movement to drain all components of the water purification unit.

4.6.2.3.3 Criteria for failure. Evidence of any one of the following conditions shall constitute failure of this test:

a. Leaks at any welded seam or any other connection.
b. Lights or any switch fails to operate.
c. Valves fail to operate properly.
d. Mechanical or electrical failure of any component or part.

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e. Failure of the unit to exhibit total dissolved solids rejection rate specified in 3.4.1.
f. Failure of unit to exhibit production rate specified in 3.4.

4.6.2.4 Operational test (6 hours). The water purification unit shall be operated as specified in 4.6.2.2 and 4.6.2.3.1 through 4.6.2.3.3 for not less than 6 hours, except condition e of 4.6.2.3.3 shall not be a criterion for failure of the 6-hour test.

4.6.2.5 Environmental.

4.6.2.5.1 High temperature. Water purification units shall be tested in accordance with MIL-STD-810, method 501, procedures I and II, category Basic Hot. Operation shall be in accordance with 4.6.2.4; Nonconformance to 3.5.1 or 3.5.2 shall constitute failure of this test.

4.6.2.5.2 Low temperature. Water purification units shall be tested in accordance with MIL-STD-810, method 502, procedures I and II, category Basic Cold. Adequate shelter will be added to maintain the water purification unit's temperature at 35 ±2 "F. Operation shall be in accordance with 4.6.2.4. Nonconformance to 3.5.1 and 3.5.2 shall constitute failure of this test.

4.6.2.6 Safety. Safety evaluation shall be done throughout testing. Nonconformance to 3.6 shall constitute failure of this test.

4.6.2.7 Human factors. The water purification unit shall be evaluated to determine its human factor characteristics. Nonconformance to 3.7 shall constitute failure of this test.

4.6.2.8 Noise level. Measure the sound level at the operator's station in accordance with MIL-STD-1474 for steady-state noise while the unit is operating at rated capacity on sea water. The generator shall be on while the sound is measured. The results shall be reported in the format required by MIL-STD-1474, figure 11. Noise contours shall be taken at not fewer than 12 equal horizontal arc increments. One increment shall be taken at the noisiest position. Noise at the operator's position shall be provided. Nonconformance to 3.7.1 shall constitute failure of this test.

4.6.2.9 Reliability. Using the MTBF specified in 3.8, the water purification unit shall be tested as specified in 4.6.2.3.1 through 4.6.2.3.3 for 1,950 hours. At least five hundred hours shall be on salt water as specified in 4.6.2.3.1. The remaining hours may be on chlorine-free fresh water. If five or fewer failures occur the unit shall be considered acceptable. (A failure is any malfunction which causes or may cause inability to

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commence operation, cessation of operation, degradation of performance capability below designated levels, or serious personnel hazards. Any malfunction which the operator can remedy will not be counted provided the repair is authorized or prescribed as an operator function and can be accomplished within 60 minutes using only controls and authorized tools or spare parts incorporated in or carried with the water purification unit. Simultaneous related failures are considered one failure. Generator failures shall not be counted.) The trailer shall exhibit no failures. If the trailer exhibits one or more failure, the entire water purification unit shall fail the test.

4.6.2.10 Maintainability. Maintenance during testing shall be in accordance with the schedule required by 3.9. The maintainability ration shall be calculated as specified by 3.9. Nonconformance to 3.9 shall constitute failure of this test.

4.6.2.11 Electromagnetic testing. The water purification unit shall be tested to determine conformance to 3.10. Nonconformance to 3.10 shall constitute failure of this test.

4.6.2.12 Transportability. Water purification units shall undergo any or all of the following transportability tests, as specified (see 6.2 and 6.15).

4.6.2.12.1 Tiedown Provisions. The tiedown provisions shall be tested in accordance with MIL-STD-209 to demonstrate conformance to 3.20.1. Any weld failure, permanent deformation, cracking, loosening, or breaking of the tiedown provision or its connecting structural components shall be cause for failure of this test.

4.6.2.12.2 Slinging Provisions. The slinging provisions shall be tested in accordance with MIL-STD-209 to demonstrate conformance to 3.20.2. Any weld failure, permanent deformation, cracking, loosening, or breaking of the slinging provision or its connecting structural components shall be cause for failure of this test.

4.6.2.12.3 Air transportability and airdrop. The water purification unit shall be tested for air transportability and airdrop in accordance with MIL-STD-1791. Inability to meet the requirements of 3.20.3 shall be cause for failure of this test.

4.6.2.12.4 Rail transportability. The first article water purification unit shall be tested in accordance with MIL-STD-810, method 516, procedure VIII. Tiedowns and bracing shall be approved by the Military Traffic Management Command Transportation Engineering Agency (MTMCTEA) before the test starts (see 6.15). Failure or permanent deformation of any component shall be cause for failure of this test.

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4.6.2.12.5 Highway transportability The first article water purification unit shall be tested in accordance with MIL-STD-810, method 514, procedure II. Trailer-mounted water purification units shall not be mounted on another trailer. Tiedowns and bracing shall be approved by the Military Traffic Management Command Transportation Engineering Agency before the test starts (see 6.15). Failure or permanent deformation of any component shall be cause for failure of this test.

4.6.2.13 Trailer electrical system. Connect the intervehicular cable to a 24-volt power source. All lights shall be checked for proper operation. Nonconformance to 3.11 shall constitute failure of this test.

4.6.2.14 Trailer brake system. When the trailer is disconnected from an appropriate air source the brakes shall engage and maintain application for not less than 15 minutes. When air in excess of 50 pounds per square inch (gauge) is applied to the emergency intervehicular hose the brakes shall release. When air is applied to the modulating intervehicular hose the brakes shall engage and hold as long as pressure is maintained. Nonconformance to 3.12 shall constitute failure of this test.

4.6.2.15 Levelling jacks. Each jack shall be individually lowered and raised to support the trailer. Nonconformance to 3.13 shall constitute failure of this test.

4.7 Inspection of packaging.

4.7.1 First article pack inspection. The first article pack shall be examined for defects specified in 4.7.2.3. Presence of one or more defects shall be cause for rejection.

4.7.2 Quality conformance inspection of pack.

4.7.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.7.2.2 Sampling. Sample size shall be determined by table I and table IIa of MIL-STD-105. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.7.2.3 Examination. Samples selected in accordance with 4.7.2.2 shall be examined for the following defects.

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138. Materials, methods, or containers not as specified for level A. Each incorrect material, method, or container shall constitute one defect.
139. Test reverse osmosis elements not removed from the unit, the unit not drained and valves left open as specified (see 5.2.1.1).
140. Caps not installed in water inlets and outlets (see 5.2.1.1).
141. Hardware items requiring a contact preservative not coated as specified (see 5.2.1.2).
142. Booster, RO, and chemical feed pumps not preserved as specified (see 5.2.1.3).
143. Ground rods not secured in the brackets (see 5.2.1.4).
144. Trailer chassis not preserved as specified (see 5.2.1.5).
145* Raw water, backwash, and distribution pumps not preserved as specified.
146. Canvas hoses not coiled and secured as specified and hard rubber hoses not left straight and secured as specified (see 5.2.1.6).
147. Technical publications not preserved as specified (see 5.2.1.7).
148. Hand tools not preserved as specified (see 5.2.1.8).
149. Cotton duck case not preserved as specified (see 5.2.1.9).
150. Extension light cord not coiled or hanked and secured as specified (see 5.2.1.10).
151. Adapters, nipples, bushings, nozzles, and strainer not wrapped with barrier material as specified (see 5.2.1.11 and 5.2.1.12).
152. Box closure for the comparator, conductivity meter, and gate valves not method V as specified (see 5.2.1.13, 5.2.1.14 and 5.2.1.16).
153. Flashlight not preserved as specified (see 5.2.1.15).
154. Elements not preserved as specified (see 5.2.1.17).
155. Box of lamps not preserved as specified (see 5.2.1.18).
156. Cargo straps not preserved as specified (see 5.2.1.19).
157. Filter tubes not installed as specified (see 5.2.1.20).
158. Components not specifically preserved, not preserved as specified for components of similar design or construction as specified (see 5.2.1.21).
159. Consolidation of components and accessories in the chests not as specified (see 5.2.1.22.1).
160. Elements not consolidated together as specified (see 5.2.1.22.2).
161. Calcium hypochlorite not consolidated as specified (see 5.2.1.22.3).
162. Distribution pump for skid mounted units not boxed as specified (see 5.2.1.22.4).
163. Other components not consolidated as specified (5.2.1.22.6).

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164. Presentation and packing not in accordance with ASTM D3958 (see 5.2.2).
165. Trailer mounted units not fitted with box type cover and the cover not secured as specified.
166. Skid mounted units not packed as specified.
167. Marking missing, illegible, incorrect? or incomplete for level A or commercial (see 5.4).

5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination within the time frame specified (see 6.2) to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3) . The first article pack may be prepared using either the first article model or a production water purification unit. When the first article model is used, any preservation or packing shall be removed by the contractor, at no cost to the Government, when requested by the Government to facilitate comparison between the first article model and production water purification units.

5.2 Preservation. Preservation shall be level A or commercial as specified (see 6.2) .

5.2.1 Level A.

5.2.1.1 Test elements. Elements used for test purposes shall be removed from the unit, the unit drained, and the drain valves left open. Install caps on all water inlets and outlets.

5.2.1.2 Hardware items. Hardware items such as bolts, nuts, linkages, pins, grease fittings, and capscrews requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P1 or P19 preservative. The preservative shall conform to the applicable specification listed in and shall be applied in conformance with MIL-P-116.

5.2.1.3 Booster pump. RO pump, and chemical feed pump. The booster pump, RO pump, and chemical feed pump shall be preserved win accordance with MIL-P-10603, level A. The RO pump housing and chemical feed pump housing shall be filled to the operating level with operating oil to which has been added 4 ounces per gallon of rust preventive in accordance with MIL-P-116, type P-20, grade optional. The pumps shall be actuated to assure coating of all parts and surfaces within the housing.

5.2.1.4 Ground rods. Ground rods shall be secured in the brackets provided on the unit.

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5.2.1.5 Trailer chassis. The components of the trailer chassis shall be preserved in accordance with MIL-V-62038, level A (mobile).

5.2.1.6 Hose assemblies. Canvas hoses shall be coiled in a minimum size coil. The coils shall be secured with cotton tape or twine. Hard rubber hoses shall be left straight for consolidation on the trailer.

5.2.1.7 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.2.1.8 Hand tools. Hand tools shall be preserved in accordance with PPP-P-40, level A.

5.2.1.9 Cotton duck case. The cotton duck case shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.2.1.10 Extension light. The cord shall be coiled or hanked to occupy minimum space and shall be secured with cotton tape or twine.

5.2.1.11 Adapters, nipples, and bushings. Each unassembled adapter, nipple, or bushing shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3 or wrapped with barrier material conforming to PPP-B-1055, class B-2, or MIL-B-121, type I, grade A, class 2. The barrier material shall be secured with tape conforming to PPP-T-76.

5.2.1.12 Nozzle and strainer. The nozzle and strainer shall each be preserved in accordance with MIL-P-116, method IC-1 or XC-3 or wrapped with barrier material conforming to PPP-B-1055, class B-2, or MIL-B-121, type I, grade A, class 2. The barrier material shall be secured with tape conforming to PPP-T-76.

5.2.1.13 Comparator. The components of the color comparator shall be placed in the case provided. The case containing the components shall be placed in a close-fitting box conforming to PPP-B-636, W6C, style optional. Box closure shall be in accordance with the appendix to the box specification, method V.

5.2.1.14 Conductivity meter. The conductivity meter shall be preserved in accordance with MIL-P-116, method llb. The inner container shall be the vendor’s carton or a box conforming to PPP-B-636, type CF, class domestic, variety SW, grade 125, style optional. The outer container shall be a box conforming to PPP-B-636, W6C, style optional. Box closure for the outer container shall be in accordance with the appendix to the box specification, method V.

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5.2.1.15 Flashlight. The flashlight shall be preserved in accordance with MIL-P-116, method IC-2, using the vendor's carton or a box conforming to PPP-B-636, class domestic, variety SW, grade 125, style optional.

5.2.1.16 Gate valves. Each gate valve shall be placed and secured in a close-fitting box conforming to PPP-B-636, W6C, style RSC. Box closure shall be in accordance with the appendix. to the box specification, method V.

5.2.1.17 Elements. Elements shall be preserved in accordance with the requirements of the applicable drawing.

5.2.1.18 Box of lamps. The lamps shall be preserved in accordance with MIL-P-116, method IC-2, using the box provided as the container.

5.2.1.19 Cargo straps. Cargo straps not used for securing equipment in position shall be preserved in accordance with MIL-P-116, method XC-1 or IC-3.

5.2.1.20 Filter tubes. one new set of filter tubes shall be installed in the cartridge filter upon completion of testing. Another set (8 each) shall be placed in the storage box on the water purification unit.

5.2.1.21 Other components. Other components not specifically mentioned herein requiring protection from corrosion or physical or mechanical damage shall be preserved as specified for components of similar design or construction.

5.2.1.22 Consolidation.

5.2.1.22.1 Chests 1 and 2. All components and accessories intended for the chests shall be placed in the applicable chest in accordance with the chest loading plan and the top closed and secured.

5.2.1.22.2 Elements. The preserved elements shall be consolidated together in a close-fitting box conforming to PPP-B-601, overseas type, style optional. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.1.22.3 Calcium hypochlorite. The containers of calcium hypochlorite shall be consolidated together in a close-fitting box and the box closed and strapped as specified in 5.2.1.22.2.

5.2.1.22.4 Distribution tromp. The distribution pump for type II or III skid mounted units shall be placed and secured in a close-fitting box and the box closed and strapped as specified in 5.2.1.23.2.

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5.2.1.22.5 Chemical feed cans and rack. The chemical feed cans shall be placed in their rack. The assembly shall then be placed and secured in a box conforming to PPP-B-601, overseas type, style optional. The box shall be closed and strapped as specified in 5.2.1.22.2.

5.2.1.22.6 Other components. Other loose components shall be placed and secured on the unit as provided for, or if more convenient to accomplish the packing specified in 5.3 shall be consolidated together in a minimum number of close-fitting boxes as specified in 5.2.1.23.2. For trailer mounted units, the boxes shall be of a size and configuration best suited to secure on the unit in a manner that will not interfere with towing or lifting.

5.2.2 Commercial. Each water purification unit, components, accessories, and support equipment shall be preserved in accordance with ASTM D3951.

5.3 Packing. Packing shall be level A or commercial, as specified (see 6.2) .

5.3.1 Level A.

5.3.1.1 Type I. Each complete type I (trailer mounted) water purification unit with all boxed or unboxed components, accessories, and support equipment secured to the unit by the provided holddowns or with strapping conforming to ASTM D3953, type 1 or 2, finish A, size as applicable, shall be packed for mobile shipment. The basic unit shall be protected with a box type cover consisting of ends, sides, and top constructed in accordance with PPP-B-601 for an overseas type, style I box. The box cover shall be the minimum size required to fit over and completely enclose the water purification structure and components extending above the trailer chassis. The plywood shall be 3/8-inch thick. The sides of the cover shall be cut to follow the contour of the fender well area and shall not interfere with lifting or towing the unit. A separate box cover constructed as specified herein for the basic unit shall be provided over the distribution pump. The box covers shall be secured in place by the strapping specified herein, wood members, nails, screws, nuts and bolts, or any combination thereof.

5.3.1.2 Type II or III. Each complete type II or III (skid mounted) water purification unit, except the elements, calcium hypochlorite, chemical feed cans and rack, and distribution pump (see 5.3.1.2.1), shall be packed in a crate conforming to MIL-C-104, type I, class 2, style a. Blocking, bracing, and anchoring to prevent movement or damage of the contents shall be in accordance with the appendix to the crate specification.

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5.3.1.2.1 Elements, calcium hypochlorite, chemical feed cans and rack, and distribution pump. The preserved and consolidated elements, calcium hypochlorite, chemical feed cans and rack, and distribution pump shall not require additional packing but shall be shipped separate from but at the same time as the water purification unit for which they are intended.

5.3.2 Commercial. Each complete water purification unit with accessories and support equipment shall be packed in accordance with ASTM D3951.

5.4 Marking.

5.4.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.4.2 Commercial. Marking for commercial packaging shall be in accordance with ASTM D3951. Weight and cube shall be marked on the shipping container.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The water purification unit is intended to supply potable water from any fresh, brackish, or sea water source.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type of water purification unit requires (see 1.2).
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents references (see 2.1.1 and 2.2).
d. When a first article is not required for or the number of units required (see 3.2).
e. Color (see 3.19).
f. Government-furnished property (see 3.21).
g. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which shall be conducted by the Government and which by the contractor (see 4.1) .
h. Time frame required for submission of the first article pack (see 5.1).
i. Degree of presentation and degree of packing required (see 5.2 and 5.3).
j. When a transportability report is required (see 6.3).

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6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Description (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/ . provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD Far Supplement 27.475-1 exempts the requirements for a DD Form 1423.

Reference paragraph	DID Number	DID Title
3.23.4.1	DI-H-24029	Certification, Welder Qualification.
3.20.4	DI-PACK-80880	Transportability Report

The above DIDs were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) , must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 First article. When a first article inspection is required, the items should be preproduction models. The first article should consist of two or more units. The contracting officer should include specific instruction in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first articles.

6.4.1 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor’s obligation to preserve, package, pack, and mark the water purification units in accordance with this specification.

6.5 Medical advisor. The Surgeon General of the Army will act as advisor to make any necessary determination on health hazards. Inquiries should be forwarded to Headquarters, Department of the Army, ATTN: DASG-TSP-E, Washington, DC, 20310.

6.6 Noise limits. When the noise level of the water purification unit exceeds the limits of MIL-STD-1474, Category D, the appropriate discussion of the noise hazard in accordance with MIL-STD-1474 (Manuals paragraph) shall include: (a) requirement for hearing protection: (b) noise level of the water purification unit; (c) distance from the water purification unit at which 85

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Db(A) will always be met; and (d) what operational situation or system configuration will reduce the level to Category D limits.

6.7 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the water purification unit covered by this specification as contained in the Federal Supply Catalogue, Department of Defense Section, Identification List C9100-IL, for FSC Group 91.

6.8 Vehicle weight classification. The contracting officer should arrange to furnish vehicle classification form to the contractor and request the return of the completed forms to the Government. The information thus obtained should be used to determine the weight classification number(s) for the water purification unit. The contractor should then be requested by the contracting officer to apply the weight classification number(s) to the water purification unit. A Government-furnished sign kit conforming to MIL-S-40626 should be used where applicable (see 3.22).

6.9 Quality assurance provisions (QAP) . The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.10 Definitions.

6.10.1 Quality assurance provisions. A QAP is a contractual requirement that supplements section 4 of the specification. QAP indicated the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.10.2 End item final inspection requirement (EIFIR). An EIFIR is a check list of quality characteristics prepared by the Government which the contractor must follow during his final inspection of the completed end item to verify complete and functional conformance to contract requirements prior to submission for final acceptance.

6.11 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional manuals, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each water purification unit.

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6.12 Provisioning. The contracting officer should include provisioning requirements for such repair parts and maintenance tools as necessary (including any special tools) , and instructions on shipment of water purification units.

6.13 Ancillary equipment. The ancillary equipment shall consist of all items listed in the latest issue of SC 4610-97CL-E16 (see 4.6.2.3.2 and 4.6.2.9).

6.14 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.21.

6.15 Rail testing. The contractor must ensure that the tiedown provisions for the rail testing are submitted to MTMCTEA in time to allow adequate review and approval. In addition, at least one week before rail testing begins (see 4.6.2.12.4), the contracting officer or the Government test engineer should ensure that MTMCTEA has been informed of the test date and location so they may witness the test.

6.16 Decision flow chart. This decision flow chart is to be used only to determine those malfunctions which will be considered in calculation of MTBF and MR (see 3.8 and 3.9).

Decision Flow Chart

Sequence	Question		Classification
1.	Does the incident concern RAM?	No	Non-RAM oriented No Score
	Yes		
2.	Does the incident concern the power source (commercial or generator)?	Yes	No Score
	No		
3.	Was incident detected during initial inspection?	Yes	No Score
	No		
4.	Did incident result from test item abuse, unrealistic operating conditions, accident, or improper maintenance or operating procedures?	Yes	No Score
	No		

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5.	Was the incident a kit installation authorized modification, engineer evaluation, or Manual/Maintenance evaluation?	Yes	No Score
		No	
6.	Was the incident detected during an inspection which required only scheduled crew maintenance, if any, or correction?	Yes	No Score
		No	
7.	Was the incident a scheduled service or replacement?	Yes	Scheduled Maintenance No Score
		No	
8.	Was the incident caused by another incident?	Yes	Unscheduled Maintenance No Score
		No	
9.	Was the incident as incipient malfunction detected during scheduled maintenance and corrected at that level?	Yes	Unscheduled Maintenance No Score
		No	
10.	Was the incident as incipient malfunction detected during operation for which correction can be deferred to the next scheduled maintenance and corrected at that level?	Yes	Unscheduled Maintenance No Score
		No	
11.	Was the incident, or could it have been, corrected by the operator within 60 minutes using on equipment maintenance tools and spare parts?	Yes	Unscheduled Maintenance No Score
		No	
12 l	Is the incident an actual malfunction for which maintenance can be deferred to the next scheduled maintenance and corrected at that level or deferred to the end of the test?	Yes	Unscheduled Maintenance No Score
		No	

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13.	Could a ten-hour work shift be completed without any degradation to operational mode mission essential functions or critical or catastrophic hazard to personnel on equipment? No Failure Unscheduled Maintenance	Yes	Unscheduled Maintenance No Score

6.17 Subject term (key word) listing.

Water, brackish
Water, contaminated
Water, drinkable
Water, fresh
Water, potable
Water, salt

6.18 Changes from Previous issue. Asterisks are not used in this revision to identify changes from the previous revision due to the extensiveness of the changes.

Custodian:
Army - ME

Review activities:
Army - MD, TE

Preparing activity:
Army - ME

Project 4610-A132

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-W-0052960B(ME)	2. DOCUMENT DATE (YYMMDD) 940624

3. DOCUMENT TITLE
WATER PURIFICATION UNIT, REVERSE OSMOSIS, 600 GPH

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) DSN	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Carolyn B. Johnson	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3468	(2) DSN 654-3468
c. ADDRESS (Include Zip Code) ARMY BELVOIR RDE CTR ATTN SATBE TSE 10101 GRIDLEY RD STE 104 FT BELVOIR VA 22060-5818	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: DEFNS QLTY & STDZN OFC 5203 LEESBURG PIKE STE 1403 FLS CHURCH VA 22041-3466 Telephone (703) 756-2340	DSN 289-2340

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